

Installing and configuring a Second ESXi Server

“In this lab, we installed a second ESXi server to build a multi-host vSphere environment and prepare for clustering and high availability. The purpose is to have another ESXi server available so virtual machines can be migrated to the second host if the first ESXi server fails or crashes.

The lab environment consists of a physical laptop or PC running VMware Workstation. On VMware Workstation, we run ESXi Server 1, which already has Linux and Windows virtual machines. We then create ESXi Server 2, where we will later install vCenter Server. vCenter Server is used in the lab to manage the environment and high availability clustering.

For ESXi Server 2, we allocated 12 GB of RAM, 2 CPUs, and a 300 GB disk. These resources were selected because the lab planned to install vCenter Server on this ESXi host and also wanted enough capacity for migrating virtual machines.

Before installing ESXi 2, we checked the network configuration. ESXi 1 was using 192.168.19.5, so we checked that 192.168.19.6 was available. We used ping to confirm that .5 was reachable and .6 was not currently being used. Because the home lab uses DHCP, we configured ESXi 2 with a static IP address.

We created a new ESXi 2 virtual machine in VMware Workstation, assigned 12 GB of memory, and used a 300 GB virtual disk. During the ESXi 6.5 installation, we accepted the license agreement, selected the 300 GB disk, created the root password, and completed the installation.

After the installation and reboot, the system initially received 192.168.19.6 through DHCP. We then accessed the ESXi DCUI by pressing F2 and configured the management network with a static IPv4 address of 192.168.19.6. We kept the existing subnet mask and default gateway.

Next, we configured the DNS settings and changed the hostname to ESXi2. We applied the network changes and tested the management network to confirm connectivity.

Finally, we pinged 192.168.19.6 and accessed ESXi 2 through the VMware Host Client using a web browser. We verified the hostname, CPU, memory, and other hardware information.

This completed the installation and basic configuration of ESXi Server 2 and prepared the lab for the next steps involving vCenter Server and high-availability clustering.”

Easy version to memorize

“We installed a second ESXi server to build a multi-host environment and prepare for high availability and clustering. We created ESXi2 in VMware Workstation with 12 GB RAM, 2 CPUs, and a 300 GB disk. We installed ESXi 6.5, configured a static IP of 192.168.19.6, set the hostname to ESXi2, tested the network, and accessed the server through the VMware Host Client. The next step is to install and configure vCenter Server.”

Item	ESXi 1	ESXi 2
IP Address	192.168.19.5	192.168.19.6
Subnet Mask	255.255.255.0	255.255.255.0
Gateway	192.168.19.2	192.168.19.2
DNS	192.168.19.2	192.168.19.2
Hostname	ESXi1	ESXi2
RAM	—	12 GB
CPU	—	2 CPUs
Disk	—	300 GB

Installation flow

Check IP availability → Create ESXi2 VM → Assign resources → Install ESXi 6.5 → Reboot → F2 → Configure Management Network → Static IP → DNS/Hostname → Test network → Ping → VMware Host Client

One-sentence version

"I installed ESXi2 in VMware Workstation, assigned 12 GB RAM, 2 CPUs, and 300 GB disk, configured the static management IP 192.168.19.6 and hostname ESXi2, tested connectivity, and accessed it through the VMware Host Client."

What is VMware vCenter Server?

"VMware vCenter Server is a centralized management utility used to manage multiple ESXi hosts, virtual machines, and related components from a single location.

In simple terms, vCenter provides a single centralized dashboard where administrators can manage multiple ESXi servers and the virtual machines running on them. This makes managing a VMware environment easier than managing each ESXi host separately.

vCenter supports several important features. **HA**, or High Availability, helps provide availability when an ESXi host fails by allowing virtual machines to be restarted or moved to another available host. **DRS**, or Distributed Resource Scheduler, helps manage resources across ESXi hosts and can balance virtual machines based on available resources. **vMotion** allows a running virtual machine to be migrated from one ESXi host to another.

Other features discussed include **vDS**, or vSphere Distributed Switch, for centralized virtual networking, **vSAN** for virtualized storage, **SSO**, or Single Sign-On, for centralized authentication, **Update Manager** for managing updates and patches, **Host Profiles** for applying consistent host configurations, and **vRealize Orchestrator** for automation.

Older versions of vCenter were installed as software on a Windows Server. Newer versions are provided as a **vCenter Server Appliance**, which is a pre-packaged virtual appliance. It contains the required operating system and vCenter software and can run as a virtual machine on an ESXi host.

In this lab environment, vCenter will be used to centrally manage the ESXi hosts and support features such as high availability and virtual machine migration.

Overall, vCenter is important because it provides centralized management and additional features for managing a larger VMware environment."

Easy version to memorize

"vCenter Server is a centralized management platform for VMware environments. It allows administrators to manage multiple ESXi hosts and virtual machines from one location. It provides features such as HA, DRS, vMotion, distributed networking, virtual storage, SSO, updates, host profiles, and automation. Modern vCenter is provided as a virtual appliance that runs as a VM on an ESXi host."

Feature	Simple meaning
vCenter	Centralized management of ESXi hosts and VMs
HA	Provides availability when an ESXi host fails
DRS	Helps balance VM resources across hosts
vMotion	Migrates a running VM between ESXi hosts
vDS	Centralized virtual networking
vSAN	Virtualized storage

Feature	Simple meaning
SSO	Centralized authentication
Update Manager	Updates and patches VMware components
Host Profiles	Applies consistent host configurations
vRealize Orchestrator	Automation and orchestration

Here is the **interview-style summary** of the vCenter lecture, keeping the source's terminology and main points.

The big picture

Without vCenter:

`ESXi 1 → Manage separately`

`ESXi 2 → Manage separately`

With vCenter:

`vCenter Server`

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`ESXi 1 → Linux VM + Windows VM`

`ESXi 2 → Other VMs`

So, the main idea to remember is:

ESXi runs the virtual machines. vCenter centrally manages multiple ESXi hosts and provides additional VMware management features.

Installing vCenter Server Appliances

"To install vCenter Server Appliance, we first need to make sure the target ESXi host has enough resources. In this lab, ESXi2 has 12 GB of RAM, 2 CPUs, and 300 GB of disk space. These resources were selected to support the vCenter Appliance and the lab environment.

First, we download the vCenter Server Appliance ISO from VMware. After mounting the ISO, we can access the VCSA files, including the OVA file. One installation method is to deploy the OVA directly from the ESXi Host Client by selecting **Create/Register VM** and choosing **Deploy a virtual machine from an OVF or OVA file**. We then select the OVA, choose the datastore and network, select a deployment size, configure passwords and other settings, and complete the deployment.

The second method is to use the **vCenter Server Appliance GUI Installer** included in the mounted ISO. The installer provides options such as Install, Upgrade, Migrate, and Restore. For this lab, we select **Install**.

The vCenter deployment uses two stages. **Stage 1** deploys the vCenter Server Appliance to the target ESXi host, while **Stage 2** completes the configuration of the deployed appliance.

For this lab, we use an **Embedded Platform Services Controller** and a **Tiny** deployment size. We deploy vCenter to ESXi2 at **192.168.19.6** and use the ESXi root account for authentication.

We configure the vCenter Appliance with the following network information: IP address **192.168.19.7**, subnet mask **255.255.255.0**, gateway **192.168.19.2**, and DNS server **192.168.19.2**. Because the lab does not have a DNS server, the system name is configured using the IP address instead of an FQDN. We also enable thin disk mode so the full disk capacity is not pre-allocated immediately.

After reviewing the deployment information, we click Finish to begin **Stage 1**, which deploys the vCenter Server Appliance to ESXi2. The deployment time depends on the available system resources.”

Easy version to memorize

“To install vCenter Server Appliance, I first download and mount the VCSA ISO. There are two methods: deploy the OVA directly through the ESXi Host Client, or use the VCSA GUI Installer. In this lab, I use the GUI Installer. I select Install, choose an Embedded Platform Services Controller and a Tiny deployment, deploy it to ESXi2 at 192.168.19.6, and configure vCenter with the IP address 192.168.19.7. Stage 1 deploys the appliance, and Stage 2 completes the configuration.”

Item	Lab Configuration
Target ESXi	ESXi2
ESXi2 IP	192.168.19.6
vCenter IP	192.168.19.7
Subnet Mask	255.255.255.0
Gateway	192.168.19.2
DNS	192.168.19.2
Deployment	Tiny
PSC	Embedded
Disk Mode	Thin
System Name	IP address
Deployment	2 stages

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Two vCenter installation methods

Method 1 — Deploy OVA

Mount ISO → VCSA folder → OVA file → ESXi Host Client → Create/Register VM → Deploy OVF/OVA

Method 2 — GUI Installer

Mount ISO → vCSA-UI Installer → Run Installer → Install → Configure → Stage 1 → Stage 2

One-sentence version

“I deployed the vCenter Server Appliance to ESXi2 using the VCSA GUI Installer, selected an Embedded PSC and Tiny deployment, configured a static IP of 192.168.19.7, and started the two-stage deployment.”

Memory tip:

ESXi1 = .5 → ESXi2 = .6 → vCenter = .7.

Ways to Access VMWare vCenter Server

“There are four main ways to access VMware vCenter Server in this lab: through a web browser, through the console, through SSH, and through the vCenter Management Interface.

The first method is the **vSphere Web Client** through a browser. We open a browser and enter the vCenter IP address, which is **192.168.19.7** in this lab. The web portal provides access to the vCenter environment. The lecture demonstrates both the Flash-based client and the HTML5 client. The vSphere Web Client is the main method for managing virtual machines and ESXi hosts.

When logging into vCenter, we use the vCenter administrator account, such as administrator@vsphere.local, rather than the ESXi root account.

The second method is **console access**. Since the vCenter Server Appliance is running as a virtual machine on ESXi2, we can access its console from the ESXi Host Client. The console provides a command-line interface where we can log in with the root account and run commands. We can also enter the shell to access the Linux command line.

The third method is **SSH**. vCenter Server Appliance runs on VMware's Linux-based Photon OS, so SSH can be used for command-line access. After enabling SSH, we can connect to the vCenter IP address using a tool such as PuTTY on Windows or SSH from a terminal. From the shell, we can execute Linux commands and perform troubleshooting or administrative tasks.

The fourth method is the **vCenter Management Interface**, also called the VMware vSphere Appliance Management interface. It is accessed through a browser using the vCenter IP address together with the management port. This interface is separate from the normal vSphere Web Client and is used for managing the vCenter appliance itself.

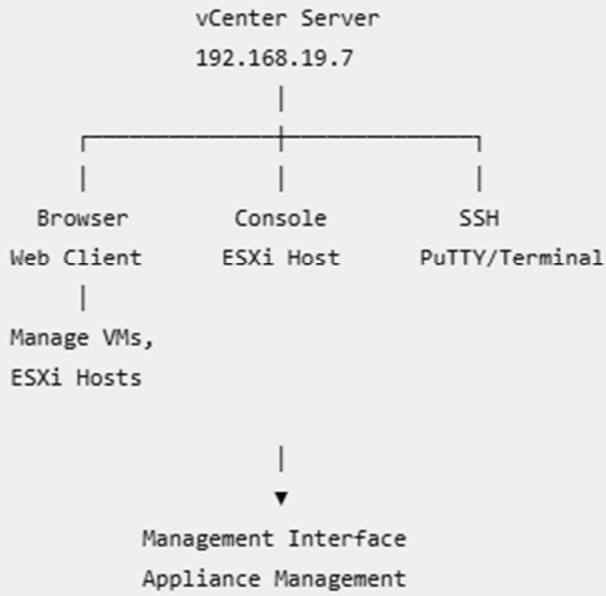
Each access method has a different purpose. The Web Client is used for normal vCenter management, while console and SSH provide command-line access. The Management Interface is used for appliance-level management.”

Easy version to memorize

“There are four ways to access vCenter: Web Client, Console, SSH, and the vCenter Management Interface. The Web Client is used to manage VMs and ESXi hosts. Console and SSH provide command-line access to the vCenter appliance. The Management Interface is used to manage the vCenter appliance itself.”

Access Method	Main Purpose	Example
Web Client	Manage VMs and ESXi hosts	Browser → 192.168.19.7
Console	Direct CLI access	ESXi Host Client → vCenter VM → Console
SSH	Remote CLI access	PuTTY / Terminal → vCenter
vCenter Management Interface	Manage the vCenter appliance	Browser → IP + management port

</> Plain text



One sentence to remember

“Web Client manages the VMware environment, Console and SSH provide CLI access, and the Management Interface manages the vCenter appliance.”

Study order: Web Client → Console → SSH → Management Interface.

vCenter Server Management Interface

“The vCenter Server Management Interface is a separate web interface used to manage the vCenter Server Appliance itself. In this lab, I access it through HTTPS using the vCenter IP address and management port **5480**, for example `https://192.168.19.7:5480`.”

After logging in with the root account, the Management Interface provides several administrative and monitoring options.

The **Summary** page shows the vCenter appliance hostname, product information, version, and overall health. It also provides information about CPU, memory, and the database. If vCenter is running slowly or experiencing problems, I can check the health status from this interface.

The interface also provides **Backup** and **Restore** functions. I can create a backup of the vCenter appliance and use that backup later if the appliance needs to be restored. It also provides **Support Bundle** functionality, which collects logs and system information that can be provided to VMware Support for troubleshooting.

From the interface, I can also **reboot or shut down** the vCenter appliance. I can check **Single Sign-On**, including the vSphere domain used for authentication.

The **Access** section allows me to check services such as SSH. The **Networking** section shows network performance and utilization. Under **Manage**, I can configure network settings such as the IP address, DNS server, and gateway.

The **Time** section is used for time and NTP configuration. **Update** is used to manage appliance updates. **Administration** provides administrative settings, while **Syslog** can be configured to send vCenter logs to a centralized logging system such as Splunk.

Finally, the **CPU and Memory** section provides performance information, and the **Database** section provides information about the embedded database.

Overall, the vCenter Management Interface is mainly used to monitor, configure, troubleshoot, update, back up, and maintain the vCenter Server Appliance itself.”

Easy version to memorize

“The vCenter Management Interface is used to manage the vCenter appliance itself. I access it through HTTPS on port 5480. I can use it to check health, configure networking, manage SSH, configure time and updates, create backups and support bundles, reboot or shut down the appliance, monitor CPU and memory, configure syslog, and check the database.”

Managing ESXi Host in vCenter

“After adding ESXi hosts to vCenter, I can manage them from a centralized location. In this lab, ESXi1 and ESXi2 are added under the **My Lab** datacenter.

When I select an ESXi host in vCenter, I have many management options. I can create a new virtual machine, deploy an OVA template, and create a resource pool. A resource pool allows me to reserve or allocate CPU and memory resources to a group of virtual machines.

I can also disconnect an ESXi host from vCenter without shutting down the host or its virtual machines. I can place the host into **Maintenance Mode** when performing maintenance. When the appropriate migration and HA features are available, virtual machines can be migrated to another host.

For power management, I can reboot or shut down the ESXi host. I can rescan storage when new storage devices are connected and configure additional virtual networking.

Another important feature is **Host Profiles**. I can capture an ESXi host's configuration and apply the same configuration to another host. This helps reduce manual configuration and provides consistency across hosts.

I can also manage VM compatibility, export system logs for troubleshooting, configure HA, assign licenses, and modify host settings such as networking, time, certificates, power management, and advanced system settings.

vCenter also provides options for moving hosts between datacenters or other organizational containers, applying tags and custom attributes, assigning permissions, and configuring alarms.

Finally, I can use orchestration features for automation and **Update Manager** to manage ESXi updates and patches.

Overall, vCenter provides centralized administration of ESXi hosts, making it easier to manage virtual machines, resources, networking, storage, maintenance, security, troubleshooting, and updates.”

Easy version to memorize

“vCenter allows me to centrally manage ESXi hosts. I can create VMs, deploy OVA templates, create resource pools, use Maintenance Mode, manage storage and networking, apply Host Profiles, configure HA, export logs, manage permissions and alarms, and update ESXi hosts.”

Option	What it does
Create New VM	Creates a virtual machine
Resource Pool	Allocates resources to VMs
Deploy OVA	Deploys a virtual appliance
Disconnect	Removes the host's connection to vCenter
Maintenance Mode	Prepares host for maintenance
Power	Reboots or shuts down the host
Storage Rescan	Detects newly attached storage
Add Networking	Configures additional virtual networks
Host Profiles	Applies a standard host configuration
Compatibility	Controls VM compatibility settings
Export System Logs	Creates logs for troubleshooting
Configure HA	Manages vSphere HA settings
Assign License	Assigns a VMware license
Settings	Configures host settings
Move To	Moves the host to another location
Tags & Attributes	Labels and identifies hosts
Permissions	Controls user access
Alarms	Provides notifications about problems
Remove from Inventory	Removes the host from vCenter inventory
Orchestrator	Automates tasks
Update Manager	Manages ESXi updates

One sentence to remember

“vCenter gives me centralized control over ESXi hosts, including VMs, resources, networking, storage, maintenance, troubleshooting, security, HA, and updates.”